



Denoising diffusion model-based subsurface modeling and quantitative interpretation

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DiffSim for geomodelling

- Diffusion models have demonstrated strong capabilities in generating realistic and diverse natural images in the computer vision domain.
- As an emerging class of deep generative methods, diffusion models are increasingly being explored for various subsurface applications.
- The proposed **DiffSim** framework generates multiple realistic geomodels that can serve as prior models for subsequent quantitative interpretation.

Training data: 2D and 3D facies models

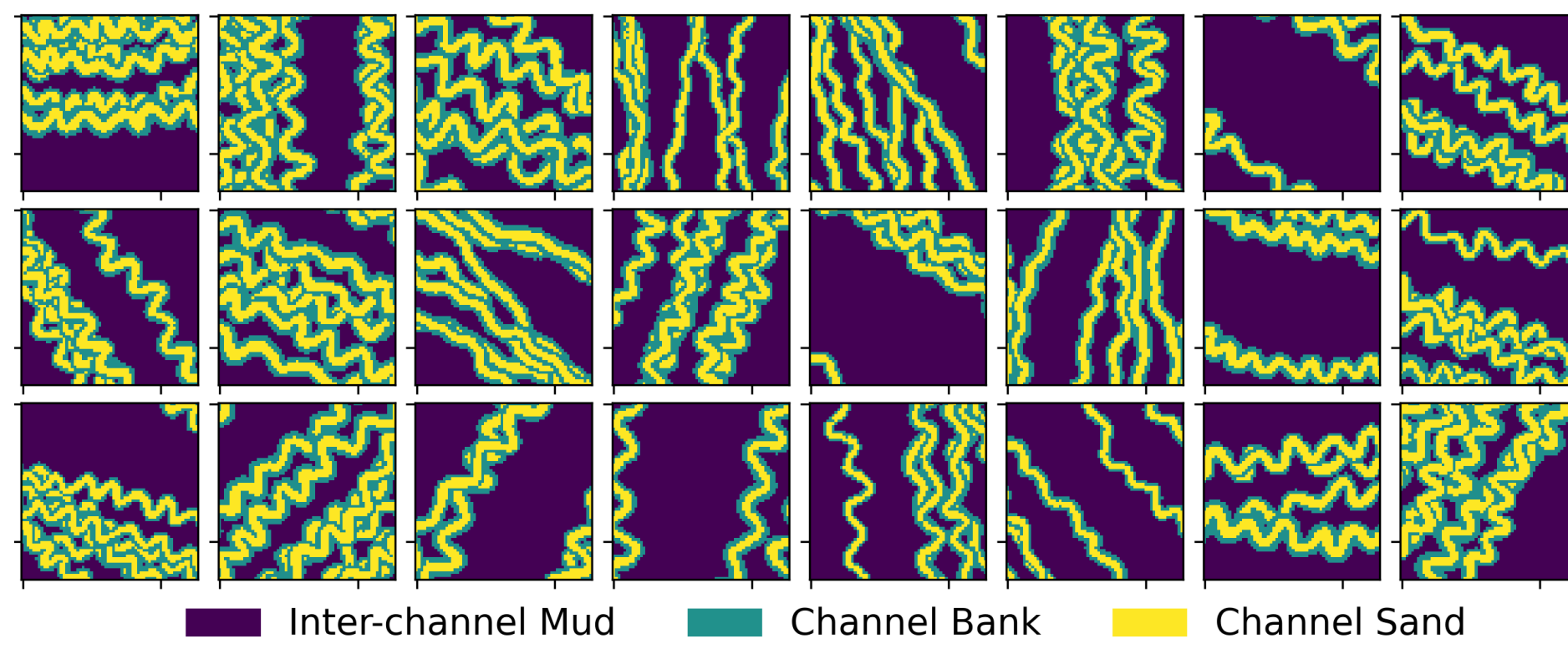


Figure 1. Some facies model examples from the first dataset.

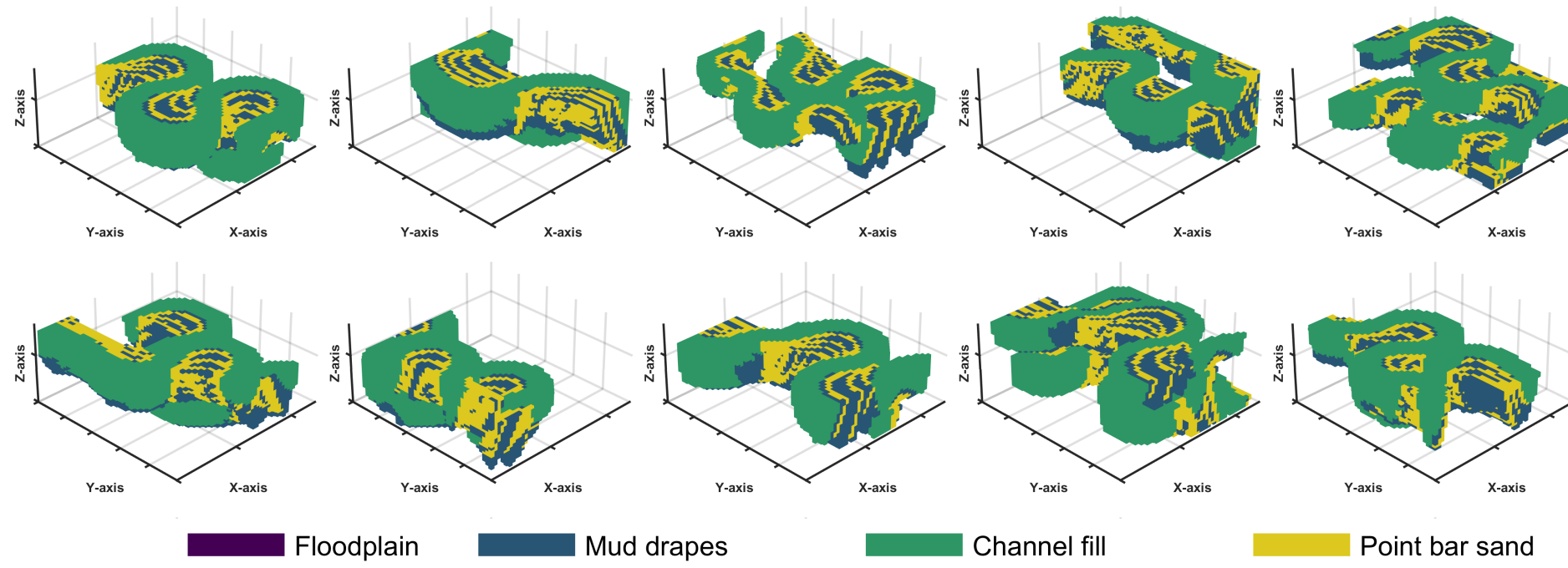
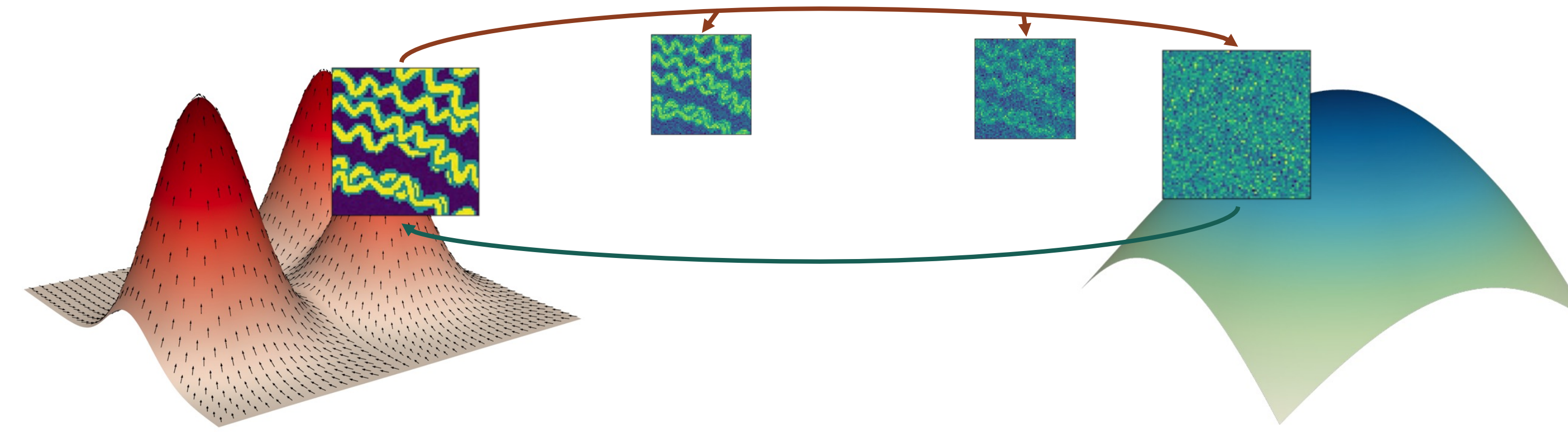


Figure 2. Samples from three-dimensional channel models.

	Training set	Test set	Model size
2D facies	32,640	3,000	64×64
3D facies	12,000	500	48×48×32

Methods: diffusion generative models (DiffSim)



$x_0 \sim p(x)$, geomodel data distribution

$x_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, standard Gaussian distribution

- Defining the transformation path
- Using neural networks (NNs) to learn the reverse path

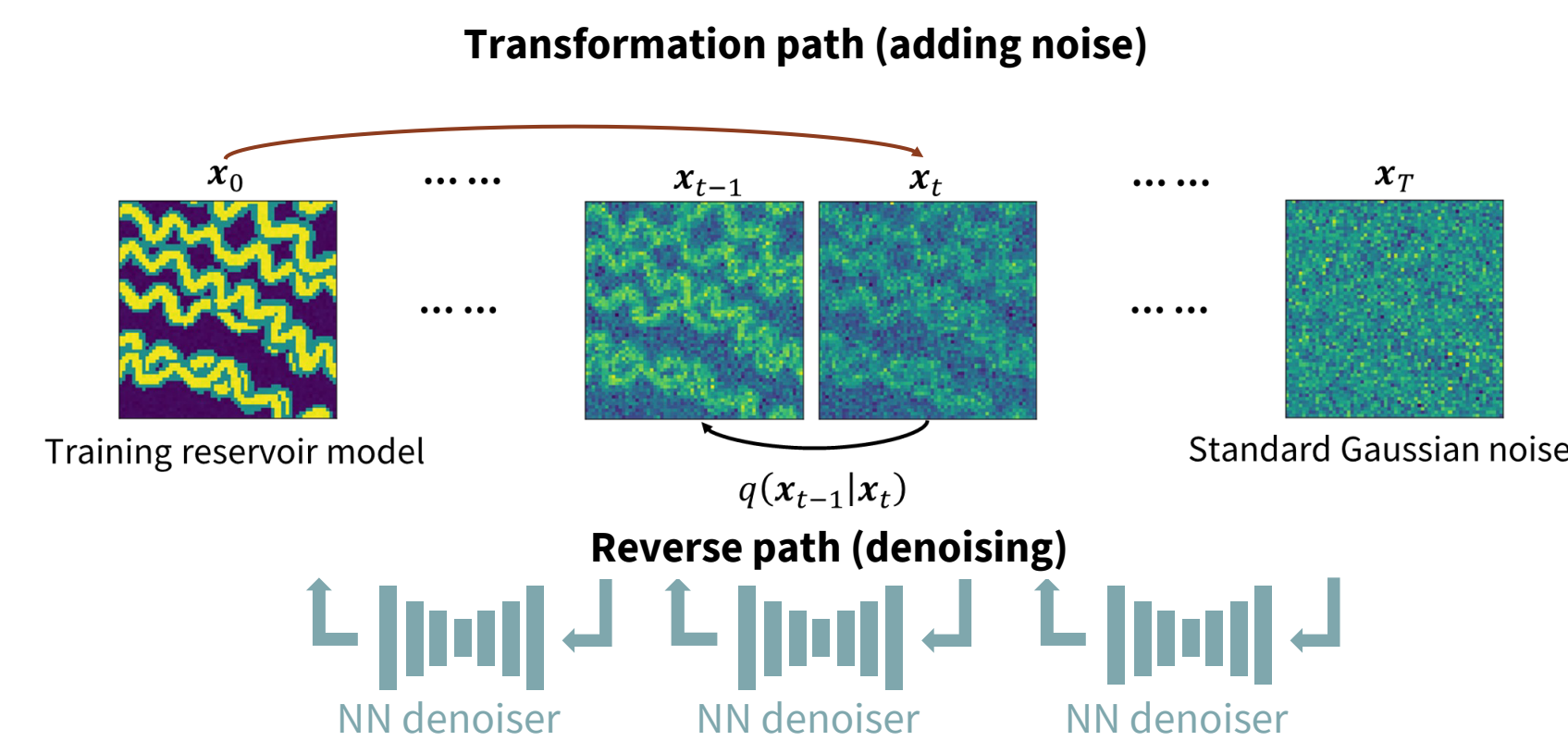
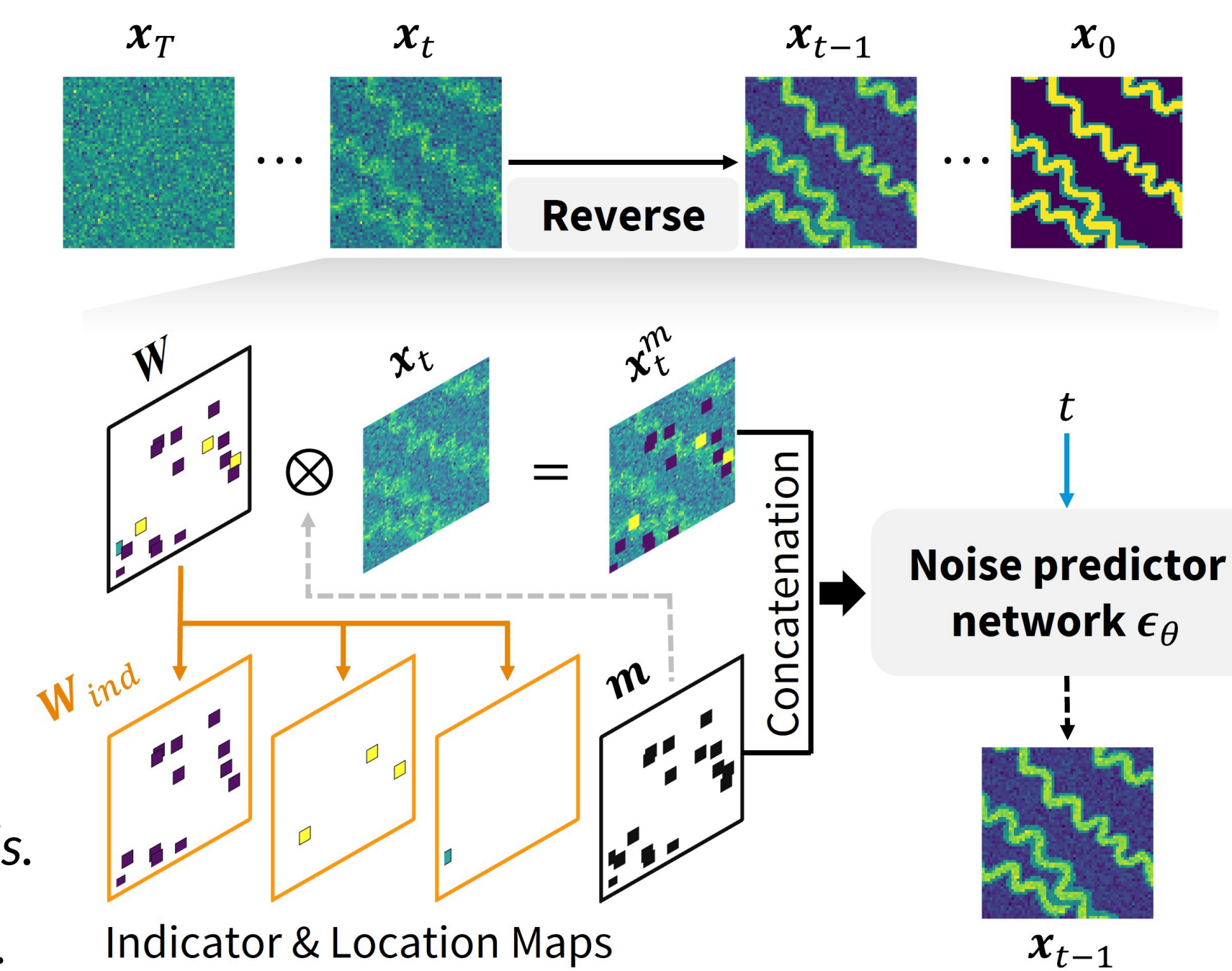


Figure 3 (top). Unconditional generation of diffusion models.

Figure 4 (right). Conditional generation of diffusion models.



Discussion & Future work

- The trained diffusion models can produce realistic and diversified facies models, demonstrating their ability to effectively capture the complex geological patterns embedded within the training conceptual geomodels.
- Future work will extend DiffSim into a Bayesian inversion framework that combines diffusion-based prior modeling with the assimilation of geophysical observations (e.g., seismic data) for quantitative interpretation and uncertainty quantification.

References

- [1] Ho, J., A. Jain, and P. Abbeel, 2020, Denoising diffusion probabilistic models: Advances in neural information processing systems, 33, 6840–6851.
- [2] Xu, M., S. Song, and T. Mukerji, 2024, DiffSim: Denoising diffusion probabilistic models for generative facies geomodeling: Fourth International Meeting for Applied Geoscience & Energy, Society of Exploration Geophysicists and American Association of Petroleum Geologists, 1660–1664.
- [3] He, K., 2024, Lecture 5: Diffusion Models — 6.S978 Deep Generative Models slides, MIT EECS. Available at https://mit-6s978.github.io/assets/pdfs/lec5_diffusion.pdf

Generated realizations conditioned on various well facies

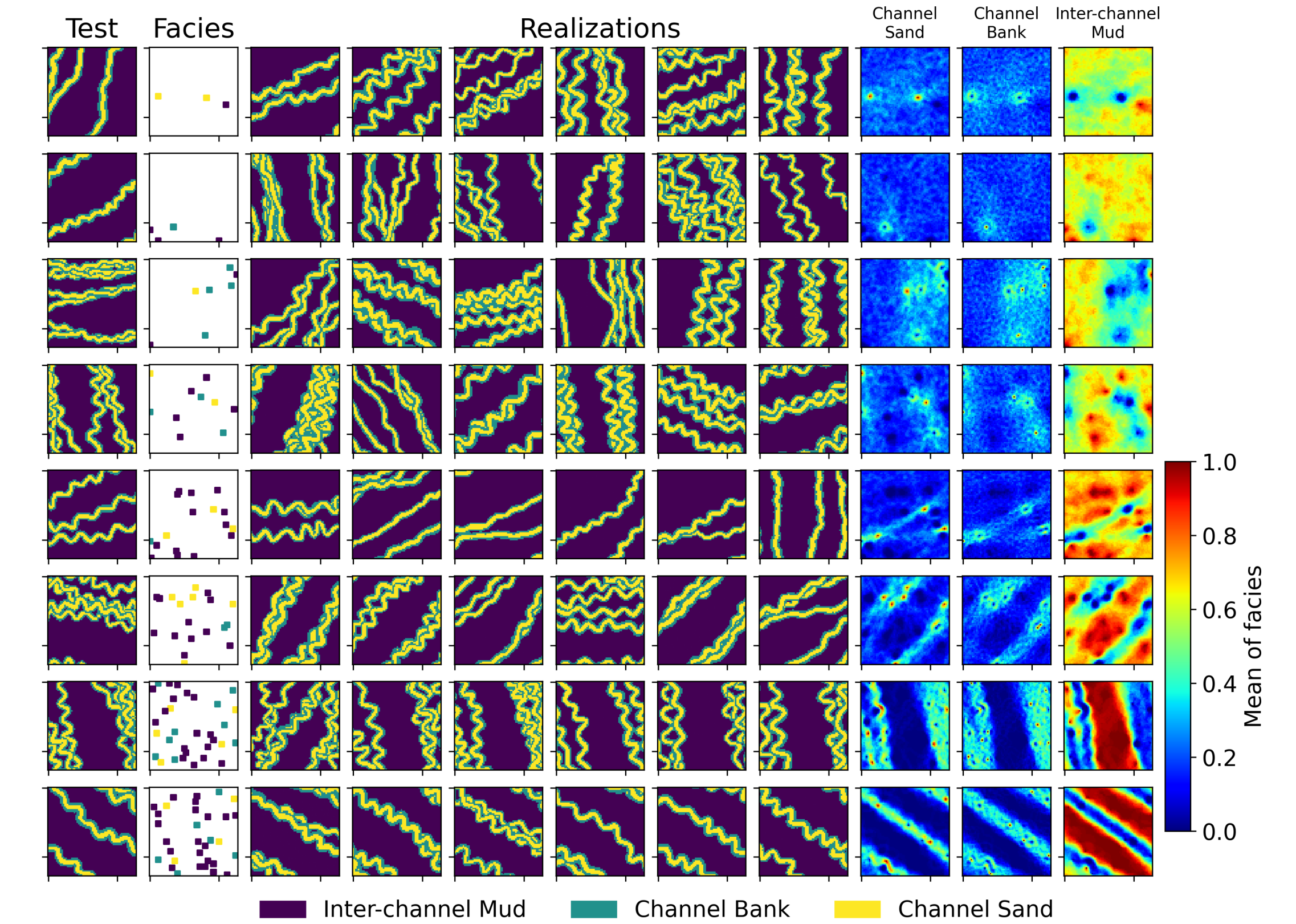


Figure 5. The generated realizations using conditional diffusion models with the different known facies as constraints. Each row includes a facies model sampled from the test set, the known facies, six out of 100 realizations, and the mean values of three facies across the 100 realizations.

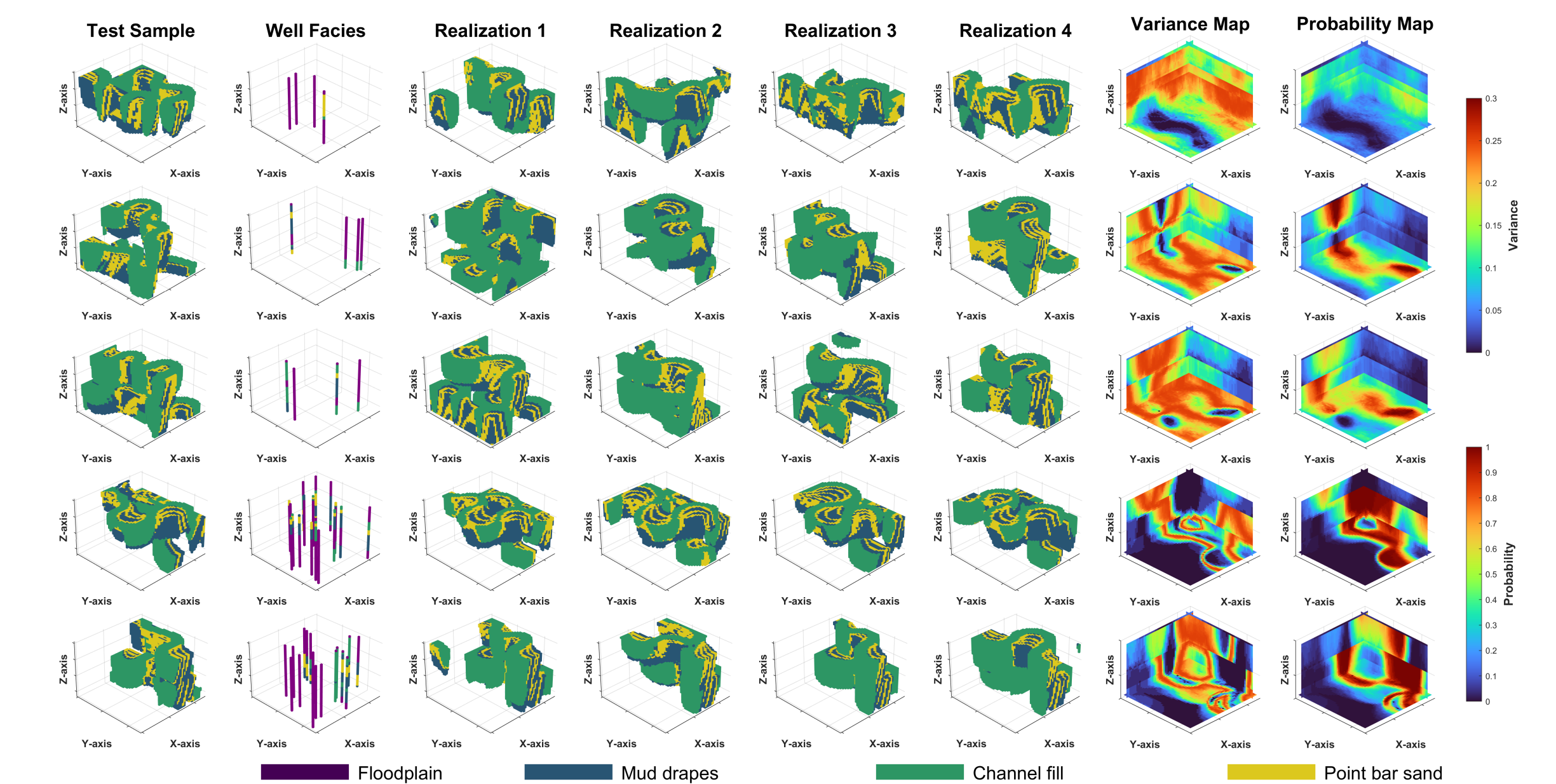


Figure 6. Conditional realizations generated from diffusion models with different known facies as constraints. Each row includes a geomodel from the test set, randomly assumed known facies, four out of 100 realizations, and generated variance and probability maps of the river facies.